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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CBCS) Degree Examinations
 December 2024 - January 2025
22EVT122 – MODELING AND ANALYSIS OF ELECTRICAL MACHINES

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Derive the Kron's primitive machine-voltage, current and power equations.	10	L*4	1	3,5
b)	A separately excited dc motor is delivering rated torque at rated speed. Find the efficiency of the motor at this operating point. The details of the machine are as follows: 1500kW,600V, rated current=2650A,600rpm, Brush voltage drop=2V, Field power input=50kW, $R_a=0.003645\Omega$, $L_a=0.1mH$, Machine frictional torque coefficient=15 N-m/(rad/sec). Field current is constant and the armature voltage is variable.	10	L3	2	3,4,5
2. a)	Derive the Mathematical model of separately excited DC motor and hence obtain the Transfer Function.	10	L3	2	3,4,5
b)	Why dynamic modeling of machine is necessary? Explain the steps involved in developing a model.	10	L2	1	3,5
3. a)	Why linearization techniques for small perturbations are necessary? Explain this technique as applied to separately excited DC motor.	10	L3	2	3,4,5
b)	Describe a space phasor and thus elaborate the phenomenon of production of a travelling MMF wave in 3phase Induction Motor.	10	L2	1	3,5
Unit – II					
4. a)	Obtain the realistic model of a single-phase transformer with off nominal turns ratio.	10	L3	4	3,5
b)	An induction motor has the following parameters: 2000hp,200V,3 phase, star connected, 4 pole, 60Hz, full load slip=0.03746. $R_s=0.02\Omega$; $R_r=0.12\Omega$; $X_m=50\Omega$; $X_{ls}=X_{lr}=0.32\Omega$ Calculate the electromagnetic torque developed by the motor when the machine is running at 1400 rpm using synchronously rotating frame theory.	10	L4	3	3,4,5
5. a)	Obtain the Electromagnetic Torque developed in 3 phase induction motor using dq reference theory.	10	L3	3	3,4,5
b)	Derive the equivalent π circuit model of a transformer in terms of admittance and off nominal turns ratio.	10	L3	4	3,5
6. a)	Obtain the per unit model of 3phase induction motor using dq modeling.	10	L3	3	3,4,5
b)	With the help of diagrams, explain the following (i) Regulating transformer for voltage control (ii) Regulating transformer for phase angle control	10	L3	4	3,5
Unit – III					
7. a)	Derive the mathematical model for a Permanent Magnet Synchronous Motor.	10	L4	5	3
b)	Discuss the method of converting synchronous machine equations in per unit.	10	L4	5	3
8. a)	Derive the mathematical model for a brushless DC Motor.	10	L4	5	3
b)	(i) Show that Park matrix is power invariant (ii) Using a power invariant transformation with θ for the transformation given by $\theta=\omega_s t + \delta$, determine f_{dq0} when $f_{abc} = \left[f_m \sin(\omega_s t + \phi), f_m \sin\left(\omega_s t + \phi - \frac{2\pi}{3}\right), f_m \sin\left(\omega_s t + \phi + \frac{2\pi}{3}\right) \right]$	10	L4	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
